

KILLARA HIGH SCHOOL



YEAR 12 MATHEMATICS ASSESSMENT 2 - 2007

TOTAL TIME: 3 hours

TOTAL MARKS: 120

Read each question thoroughly.

Attempt all questions.

Start each question on a new page.

Non-Programmable Calculator may be used.

All necessary working out must be shown to obtain full marks

Total marks 120

Attempt all questions 1 – 10

All questions are approximately of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklet available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Evaluate (round the answer off to 3 significant figures). $\sqrt{\frac{5.2^3 - 10.8}{15.9 \times 23.6}}$ 1
- (b) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2}$ 1
- (c) Solve for x: $|3x - 4| \geq 8$ 2
- (d) Prove the identity: $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$. 2
- (e) Express as a fraction in its simplest term: $\frac{5}{x^2 - 5x + 4} - \frac{3}{x^2 - 16}$ 2
- (f) Rationalise the denominator and express in simplest terms $\frac{2 - \sqrt{3}}{2\sqrt{3} + 1}$ 2
- (g) Given that $\log_a 10 = 3.321928$ and $\log_a 7 = 2.807355$, calculate $\log_a 49000$ 2

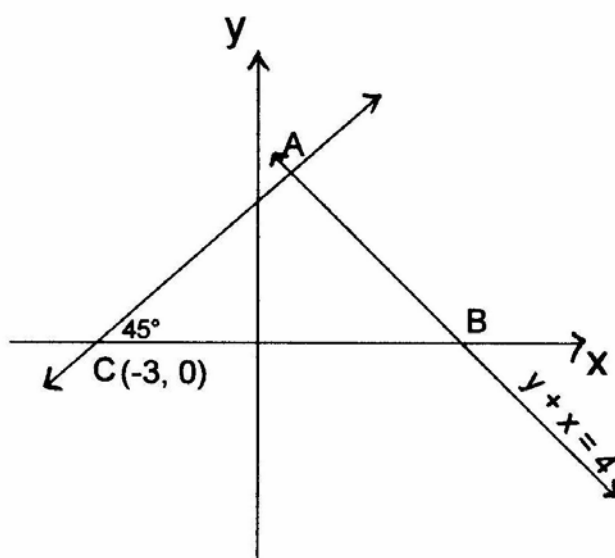
Question 2 (11 marks) Use a SEPARATE writing booklet.

- (a) Show, by shading on a sketch, the region defined by the inequalities

3

$$\begin{aligned} 2x + 3y &< 6, \\ x - y &\geq 1 \end{aligned}$$

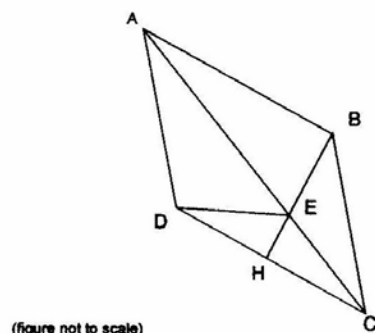
- (b) In the diagram below, the line through $C(-3, 0)$, makes an angle of 45° to the x -axis and intersects the line $y + x = 4$ at the point A . Copy the diagram into your examination booklet.



- i) Show that AC is perpendicular to AB . 1
- ii) Find the equation of the line through B which is parallel to AC . 2
- iii) Show that the equation of the line through the point C which is parallel to the line AB is $y + x + 3 = 0$. 1
- iv) The lines from part (ii) and (iii) intersect at D . Find the coordinates of the point D . 2
- v) What type of quadrilateral is $ABDC$? Give reasons. 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a)

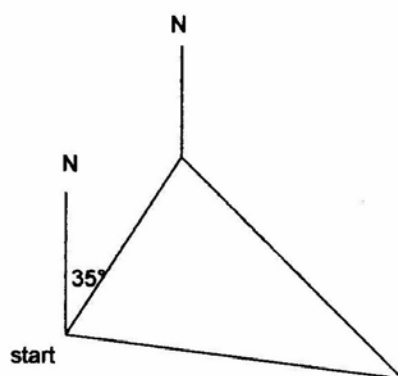


ABCD is a rhombus, BH is perpendicular to DC and intersect AC at E.

- i) Copy the diagram into your examination booklet and prove that $\triangle ADE \equiv \triangle ABE$. 3
- ii) Show that $\angle ABE$ is a right angle. 1

- (b) Solve $\cos \theta = 2 \cos^2 \theta$ for the domain $0 \leq \theta \leq 360^\circ$ 3

- (c) A helicopter flies 54 km on a bearing of 035° . It then flies 67 km on a bearing of 140° .



- i) How far is it from its starting point (correct to one decimal place)? 2
- ii) What bearing needs to be taken to return to its starting point? Give your answer correct to the nearest degree. 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a)
- i) Find the locus of a point P which moves so that $PA = \sqrt{2} PB$ where P is the point (x, y), A is (-1, 0) and B is (1, 1). 2
 - ii) Show that the locus is a circle 1
 - iii) Find the length of the radius and the coordinates of the centre of the circle. 2
- (b) A parabola has its vertex at the point (4, -3) and focus at the point (1, -3).
- i) What is the focal length? 1
 - ii) What is the equation of the directrix? 1
 - iii) What is the equation of the parabola? 2
- (c) Solve for x: $9^x - 12(3^x) + 27 = 0$ 3

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) The roots of the equation $2x^2 + bx + c = 0$ are $-\frac{1}{2}$ and -4 . Find the values of b and c . 2

- (b) For what values of m does the quadratic equation $x^2 - 2mx + 8m - 15 = 0$ have two unequal roots? 2

- (c) Find the constants a , b , and c such that $x^2 + 2x - 2 \equiv ax(x + 1) + bx^2 + c(x + 1)$. 3

- (d) A function is defined by the rule $f(x) = \begin{cases} \sqrt{x-1}, & x \geq 1 \\ x-1, & x < 1 \end{cases}$

i) Find $f(-5) - f(10)$ 1

ii) If $f(x) = 8$, find x . 2

- (e) State the largest possible domain and range for the function defined by

$$f(x) = |x| - 2 \quad 2$$

Question 6 (13 marks) Use a SEPARATE writing booklet.

(a) Differentiate with respect to x :

i) $5x^3 - \frac{2}{x} + 5$ 1

ii) $\frac{8x-1}{3x+1}$ 2

iii) $3x^2(2x+1)^4$ 2

iv) $\frac{10}{\sqrt{(12-5x)^3}}$ 2

(b) For the curve $y = x^3 + x^2 - x$

i) find the stationary points and determine their nature. 3

ii) find any points of inflexion. 1

iii) sketch the curve. 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) Given the series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$
- i) find the seventh term correct to 3 decimal places. 1
 - ii) find the sum to infinity. 1
- (b) The first term of an arithmetic series is 5, and the fifteenth term is three times the fourth term. Find the common difference. 3
- (c) Jane invests \$800 at the beginning of each year in a superannuation fund. Compound interest is paid at 12% per annum on the investment. The first \$800 is to be invested at the beginning of 2008 and the last is to be invested at the beginning of 2022.
- i) Calculate the amount to which the 2008 investment will have grown by the beginning of 2023. 2
 - ii) Show that the amount (A) to which the total investment will have grown by the beginning of 2023 is: 2
- $$A = 800 \left[\frac{1.12(1.12^{15} - 1)}{0.12} \right]$$
- iii) Find the total value of investment at the beginning of 2023. 1
- (d) Simplify $\log(x^2 - 2x + 1) - \log(x - 1)$ 2

Question 8 (13 marks) Use a SEPARATE writing booklet.

- (a) A sealed tin rectangular box is to have a square base and a volume of 64 cm^3 .
If the length of the edge of the base is $x \text{ cm}$,

i) express the height h in term of x ,

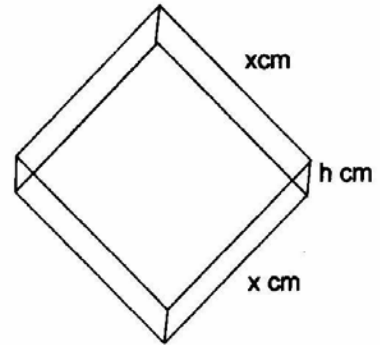
1

ii) show that the total surface area
 $y \text{ cm}^2$ is given by $y = \frac{256}{x} + 2x^2$,

2

iii) find the minimum surface area of
this box.

3



- (b) If $\frac{dy}{dx} = 2x + 3$ and $y = 5$ when $x = -2$, express y as a function of x ,

2

- (c) Find the indefinite integrals:

i) $\int (x^4 - 3x^2 + 3) dx$

1

ii) $\int \frac{4x^3 - x^2 + 5}{x^2} dx$

2

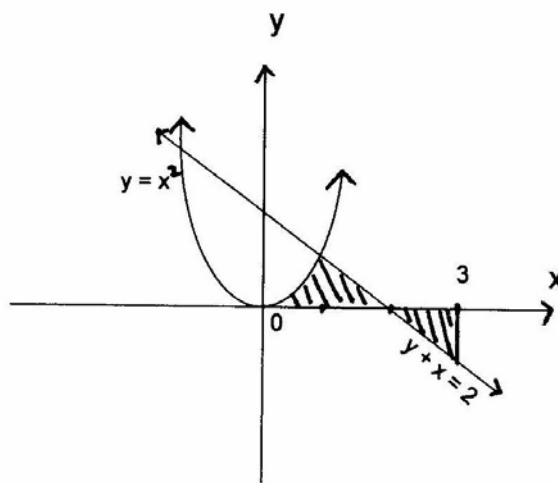
iii) $\int \frac{5}{\sqrt{(2-3x)^3}} dx$

2

Question 9 (12 marks) Use a SEPARATE writing booklet.

(a) Consider the area of the shaded region bounded by $y + x = 2$, $y = x^2$ and the x-axis as shown.

- i) Find the x-coordinates of the points of intersection of the curve $y = x^2$ and the line $y + x = 2$. 2
- ii) Find the area of the shaded region. 3



(b) Consider the function given by $y = x \log_e x$.

- i) Copy and complete the following table in your examination booklet. 1

x	1	2	3	4	5
y					

- ii) Apply Simpson's rule with five function values to find an approximation to 2

$$\int_1^5 x \log_e x \, dx$$

(c) A cone is formed by rotating the segment of the line $x + 2y = 4$ cut off by the axes about the Y-axis.

- i) Draw a diagram to illustrate this information. 1
- ii) Find the volume of the cone. 3

Question 10 (11 marks) Use a SEPARATE writing booklet.

(a) Differentiate with respect to x :

i) $\log_e 5x + e^{-3x} - 2x$ 1

ii) $\log_e (x^2 - 3x + 1)$ 1

iii) $e^{2x} \log_e x$ 2

(b) Find the indefinite integral:

i) $\int \frac{x}{x^2 + 1} dx$ 1

ii) $\int \frac{x-1}{x+4} dx$ 2

(c) Evaluate $\int_1^2 (e^{2x+3} + 1) dx$ 2

(d) Find the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 1$. 2

(Q1)

$$a) = 0.58816 \\ = 0.588$$

(1)

$$c) \quad 3x-4 \geq 8 \quad \& \quad 3x-4 \leq -8 \\ 3x \geq 12 \quad \quad \quad 3x \leq -4 \\ x \geq 4 \quad \quad \quad x \leq -\frac{4}{3}$$

(2)

$$d) \quad \text{L.H.S} = 1 - 2\tan x + \tan^2 x + 1 + 2\tan x + \tan^2 x \\ = 2 + 2\tan^2 x \\ = 2(1 + \tan^2 x) \\ = 2\sec^2 x = \text{R.H.S}$$

(2)

$$e) \quad \frac{5}{(x-4)(x-1)} - \frac{3}{(x+4)(x-4)} \\ = \frac{5(x+4) - 3(x-1)}{(x^2-16)(x-1)} = \frac{5x+20-3x+3}{(x^2-16)(x-1)} \\ = \frac{2x+23}{(x^2-16)(x-1)}$$

(1)

$$f) \quad \frac{(2-\sqrt{3})(2\sqrt{3}-1)}{12-1} = \frac{4\sqrt{3}-2-6+\sqrt{3}}{11} \\ = \frac{5\sqrt{3}-8}{11}$$

(1)

$$b) = \lim_{x \rightarrow 2} \frac{(x-1)(x-2)}{(x-2)} = 1$$

(1)

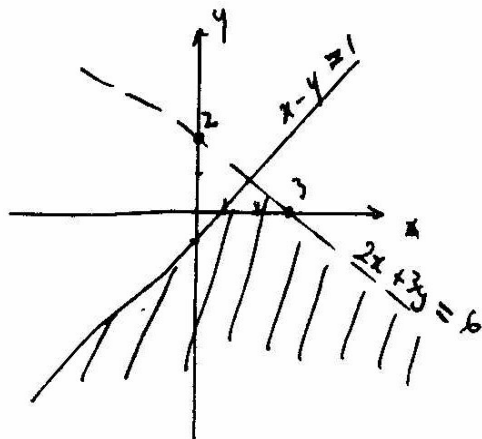
$$g) \quad \log_a 49000 = \log_a (49 \times 1000) \\ = \log_a 49 + \log_a 1000 \\ = 2\log_a 7 + 3\log_a 10 \\ = 2 \times 2.807355 + 3 \times 3.3219 = 15.58$$

(1)

(12)

[Q2]

a)



① correct { dark lines
broken

① correct shaded region

① labelling

① for any mistake.

b) (i) $m_{AC} = \tan 45^\circ$
 $= 1$ $m_{AB} = -1$

$m_{AC} \times m_{AB} = -1 \therefore AC \perp AB$ ①

i) $\frac{y-0}{x-4} = 1 \Rightarrow x-4 = y$ ② Co-ords of B = (4, 0)

ii) $\frac{y-0}{x+3} = -1 \Rightarrow x+3+y=0$ ②

v) $\begin{array}{r} x-y = +4 \\ + \\ x+y = -3 \\ \hline 2x = +1 \end{array}$

$x = +\frac{1}{2}$ Sub. into ① ①

$\Rightarrow y = -3\frac{1}{2}$ ①

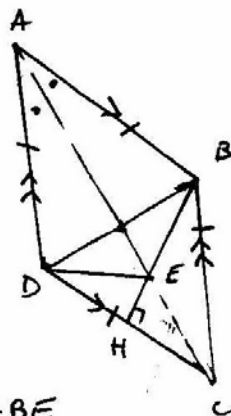
$\therefore D(+\frac{1}{2}, -3\frac{1}{2})$

vi) ABDC has opp. sides // with 1 right angle ①

$\therefore ABCD$ is a rectangle. ①

Q3

a)

i) in $\triangle ADE$ and $\triangle ABE$

$$AB = AD \text{ (sides of a rhombus)} \quad (1)$$

AE is common

$$\angle DAE = \angle EAB \text{ (Diag. of rhombus bisects the angle)} \quad (1)$$

$$\therefore \triangle ADE \equiv \triangle ABE \text{ (SAS)} \quad (1)$$

ii) $AB \parallel DC$ (opp. sides of rhombus)

$$\therefore \angle EHC = \angle ABH \text{ (alt. } \angle \text{'s in } \parallel \text{ lines)} \quad (1)$$

b)

$$\cos \theta = 2 \cos^2 \theta$$

$$2 \cos^2 \theta - \cos \theta = 0$$

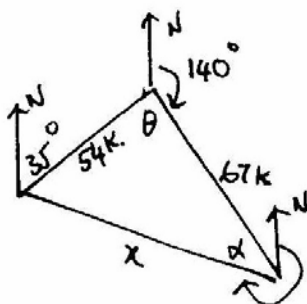
$$\cos \theta (2 \cos \theta - 1) = 0 \quad (1)$$

$$\cos \theta = 0 \Rightarrow \theta = 90^\circ, 270^\circ \quad (1)$$

$$2 \cos \theta - 1 = 0 \Rightarrow \theta = 60^\circ, 300^\circ \quad (1)$$

$$\therefore \theta = 90^\circ, 60^\circ, 270^\circ, 300^\circ$$

c)



$$i) \theta = 360^\circ - 140^\circ - (180^\circ - 35^\circ) = 75^\circ \quad (1)$$

$$x^2 = 54^2 + 67^2 - 2 \times 54 \times 67 \times \cos 75^\circ$$

$$\approx 5532.1854$$

$$x = 74.4 \text{ km} \quad (1)$$

$$ii) \frac{74.4}{\sin 75^\circ} = \frac{54}{\sin \alpha} \Rightarrow \sin \alpha = \frac{54 \times \sin 75^\circ}{74.4} = 0.201 \quad (1)$$

$$\text{Bearing} = 360^\circ - (180^\circ - 140^\circ) - 44^\circ 30' 48'' = 275^\circ \text{ T} \quad (1)$$

Q4] i) $PA = \sqrt{2} PB$
 $PA^2 = 2PB^2$

$$(x+1)^2 + y^2 = 2\{(x-1)^2 + (y-1)^2\} \quad (1)$$

$$x^2 + 2x + 1 + y^2 = 2x^2 - 4x + 2 + 2y^2 - 4y + 2$$

$$x^2 + y^2 - 6x - 4y = -3 \quad (1)$$

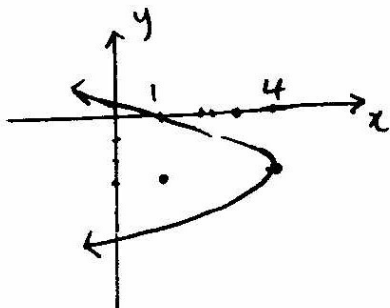
ii) $x^2 - 6x + 9 + y^2 - 4y + 4 = -3 + 9 + 4$

$$(x-3)^2 + (y-2)^2 = 10 \quad (1)$$

iii) radius = $\sqrt{10}$ (1)

Centre (3, 2) (1)

(b)



$$(y-k)^2 = -4a(x-h)$$

i) $a = 3$ (1)

ii) directrix $x = 7$ (1)

iii) $(y+3)^2 = -12(x-4)$ (2)

(c)

let $X = 3^x$

$$X^2 - 12X + 27 = 0 \quad (1)$$

$$(X-9)(X-3) = 0$$

$$3^x = 9 \quad \& \quad 3^x = 3$$

$$x = 2 \quad x = 1$$

(1)

(1)

Q5

$$a) \quad -\frac{1}{2} - 4 = -\frac{b}{2}$$

$$\frac{b}{2} = 4\frac{1}{2}$$

$$b = 9 \quad (1)$$

$$-\frac{1}{2}x - 4 = \frac{c}{2}$$

$$c = 4 \quad (1)$$

b) for 2 unequal roots $\Delta > 0$

$$4m^2 - 4 \times 1 \times (8m - 15) > 0 \quad (1)$$

$$4m^2 - 32m + 60 > 0 \Rightarrow m^2 - 8m + 15 > 0$$

$$(m-5)(m-3) > 0$$

$$\therefore m > 5 \text{ and } m < 3 \quad (1)$$

$$\begin{aligned} c) \quad ax(x+1) + bx^2 + c(x+1) &= ax^2 + ax + bx^2 + cx + c \\ &= (a+b)x^2 + (a+c)x + c \quad (1) \end{aligned}$$

$$\therefore c = -2$$

$$a+c = 2$$

$$a = 4 \quad (1)$$

$$a+b = 1$$

$$b = -3 \quad (1)$$

$$d) \text{ i) } f(-5) - f(10) = -6 - 3 = -9 \quad (1)$$

$$\text{ii) } \sqrt{x-1} = 8 \quad (1)$$

$$x-1 = 64$$

$$x = 65 \quad (1)$$

e)

domain : x is all real (1)

range : $y \geq -2 \quad (1)$

$$\boxed{6} \quad i) = 15x^2 + \frac{2}{x^2} \quad (1)$$

$$ii) \frac{8(3x+1) - 3(8x-1)}{(3x+1)^2} = \frac{24x+8-24x+3}{(3x+1)^2} = \frac{11}{(3x+1)^2} \quad (2)$$

$$\begin{aligned} iii) &= 6x(2x+1)^4 + 3x^2 \times 4 \times 2(2x+1)^3 \\ &= 6x(2x+1)^4 + 24x^2(2x+1)^3 \\ &= 6x(2x+1)^3(2x+1+4x) \\ &= 6x(2x+1)^3(6x+1) \quad (2) \end{aligned}$$

$$iv) = \frac{10}{\sqrt{(12-5x)^5}} \times -5x - \frac{3}{2} = \frac{75}{\sqrt{(12-5x)^5}} \quad (2)$$

$$b) \quad y = x^3 + x^2 - x$$

$$y' = 3x^2 + 2x - 1$$

$$i) \text{ for stationary point } y' = 0$$

$$\Rightarrow 3x^2 + 2x - 1 = 0$$

$$(x+1)(3x-1) = 0 \Rightarrow x = -1 \text{ \& } x = \frac{1}{3} \quad (1)$$

$$f''(x) = 6x + 2$$

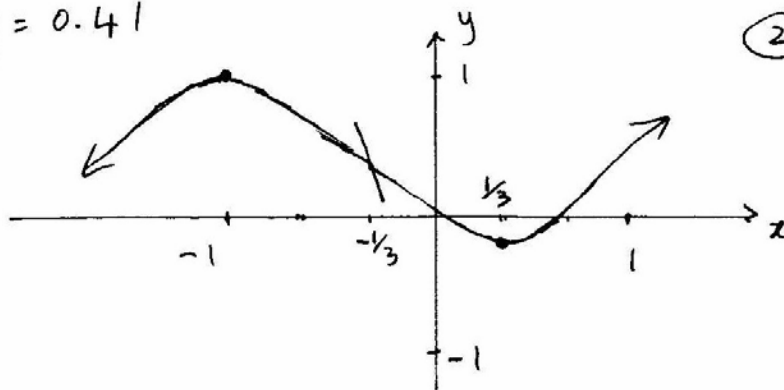
$$f''(-1) = -4 < 0 \therefore \text{max. point at } (-1, 1) \quad (1)$$

$$f''\left(\frac{1}{3}\right) = 4 > 0 \therefore \text{min. point at } \left(\frac{1}{3}, -0.185\right) \quad (1)$$

$$ii) \text{ point of inflexion } y'' = 0$$

$$6x + 2 = 0 \quad (1)$$

$$x = -\frac{1}{3}, y = 0.41 \quad (2)$$



① - correct shape

① - labelling.

Q7 i) $T_1 = 1 \times \left(\frac{1}{2}\right)^6$ ①
 $= 0.017$

ii) $S_p = \frac{a}{1-r} = \frac{1}{1+\frac{1}{2}} = \underline{\underline{\frac{2}{3}}}$ ①

⑥ $T_4 = 5 + 3d$ ① $T_{15} = 3T_4 \Rightarrow 5 + 14d = 3(5 + 3d)$ ①
 $T_{15} = 5 + 14d$
 $5d = 10$
 $\underline{\underline{d = 2}}$ ①

⑦(i) $A_1 = 800 (1.12)^{15} = \underline{\underline{\$4378.85}}$ ①

ii) $A_2 = 800 (1.12)^{14}$
 $A_3 = 800 (1.12)^{13}$
 $\vdots$
 $A_{15} = 800 (1.12)^1$ ①

Total value $A = A_{15} + \dots + A_1$
 $= 800 \times 1.12 + 800 \times 1.12^2 + \dots + 800 \times 1.12^{15}$ ①
 $= \frac{800 \times 1.12 (1.12^{15} - 1)}{1.12 - 1}$ ①
 $= \frac{800 \times 1.12 (1.12^{15} - 1)}{0.12}$

iii) $A = \$33403$ ①

⑧ $\log(x^2 - 2x + 1) - \log(x - 1) = \log \frac{x^2 - 2x + 1}{x - 1}$ ①
 $= \log \frac{(x-1)(x-1)}{x-1}$
 $= \log(x-1)$ ①

28

a) i) $x^2 \times h = 64 \Rightarrow h = \frac{64}{x^2}$ (1)

ii) S.A = $2x^2 + 4(x \times h)$ (1)

$$= 2x^2 + x \times \frac{256}{x^2}$$

$$= 2x^2 + \frac{256}{x}$$
 (1)

iii) $\frac{dA}{dx} = 4x - \frac{256}{x^2} = 0$

$$4x^3 - 256 = 0$$
 (1)

$$x^3 = 64 \Rightarrow x = 4$$
 (1)

$$\therefore S.A = 2 \times 4^2 + \frac{256}{4} = 32 + 64 = 96 \text{ cm}^2$$
 (1)

b) $y = \int 2x + 3 \, dx = x^2 + 3x + C$ (1)

$$5 = (-2)^2 - 3 \times 2 + C$$

$$C = 7$$

$$y = x^2 + 3x + 7$$
 (1)

c) i) $= \frac{x^5}{5} - x^3 + 3x + C$ (1)

ii) $\int \frac{4x^3 - x^2 + 5}{x^2} \, dx = \int 4x - 1 + \frac{5}{x^2} \, dx$ (1)

$$= 2x^2 - x - \frac{5}{x} + C$$
 (1)

iii)

$$\int \frac{5}{\sqrt{(2-3x)^3}} \, dx = \frac{+10}{3\sqrt{2-3x}} + C$$
 (2)

Q9 1

(a) i) $y = x^2$ ①
 $x + y = 2$ ② sub. ① → ②

$\Rightarrow x^2 + x = 2$ ①
 $x^2 + x - 2 = 0$
 $(x-1)(x+2) = 0 \Rightarrow x = 1, -2$ ①

(ii)

$A = \int_0^1 x^2 dx + 2 \int_1^2 2 - x dx$

$= \left[\frac{x^3}{3} \right]_0^1 + 2 \left[2x - \frac{x^2}{2} \right]_1^2 = \frac{1}{3} + 2 \left(2 - 1\frac{1}{2} \right)$
 ① ① $= 1\frac{1}{3}$ sq. units. ①

(b) i)

x	1	2	3	4	5
y	0	1.386	3.296	5.545	8.047

①

ii)

$A = \int_1^5 x \log_e x dx$

$h = \frac{5-1}{4} = 1$

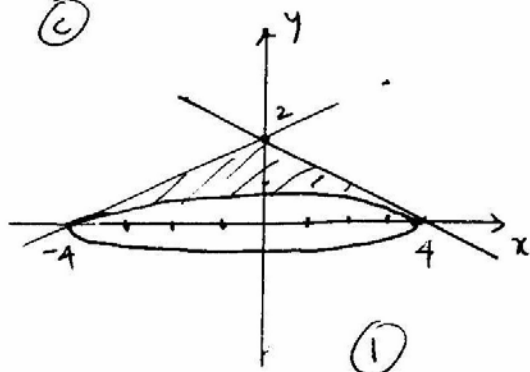
$\doteq \frac{1}{3} \left\{ f(1) + f(5) + 2 f(3) + 4(f(2) + f(4)) \right\}$

①

$\doteq \frac{1}{3} (0 + 8.047 + 2 \times 3.296 + 4 \times 1.386 + 4 \times 5.545)$

$\doteq 14.121$ sq units. ①

(c)



$x = 4 - 2y$

$x^2 = 16 - 16y + 4y^2$ ①

$V = \pi \int_0^2 16 - 16y + 4y^2 dy$

$= \pi \left[16y - 8y^2 + \frac{4}{3}y^3 \right]_0^2$ ①

⑫

$= \frac{32\pi}{3}$ or 33.51 unit³ ①

Q10

a)

$$\log_a 49000 = \log_a (49 \times 1000)$$

$$= 2\log_a 7 + 3\log_a 10$$

$$= 2 \times 2.807355 + 3 \times 3.321928$$

$$\div 15.58$$

(1)

$$a) \quad i) \quad \frac{d}{dx} \ln 5x + e^{-3x} - 2x = \frac{1}{x} - 3e^{-3x} - 2$$

(1)

$$ii) \quad \frac{d}{dx} \log_e (x^2 - 3x + 1) = \frac{2x - 3}{x^2 - 3x + 1}$$

(1)

$$iii) \quad \frac{d}{dx} e^{2x} \log_e x = 2e^{2x} \ln x + \frac{e^{2x}}{x}$$

(2)

$$b) \quad i) \quad \int \frac{x}{x^2+1} dx = \frac{1}{2} \ln(x^2+1) + C$$

(1)

$$ii) \quad \int \frac{x-1}{x+4} dx = \int \left(\frac{x+4}{x+4} - \frac{5}{x+4} \right) dx$$

(1)

$$= x - 5 \ln(x+4) + C$$

(1)

$$c) \quad \int_1^2 (e^{2x+3} + 1) dx = \left[\frac{1}{2} e^{2x+3} + x \right]_1^2$$

(1)

$$= 550.31658 - 75.20658$$

$$\div 475.11$$

(1)

d)

$$\frac{d}{dx} e^{2x} = 2e^{2x}$$

$$\text{at } \begin{cases} x=1 \\ y=e^2 \end{cases} \quad m = 2 \times e^{2 \times 1} = 2e^2$$

(1)

$$\frac{y - e^2}{x - 1} = 2e^2 \Rightarrow y - e^2 = 2e^2 x - 2e^2$$

$$y - 2e^2 x + e^2 = 0$$

or

$$y - 14.8x + 7.4 = 0$$

(1)